TORONTO METROPOLITAN UNIVERSITY

ZERO TRUST NETWORK ARCHITECTURE

Prevention, Detection and Automated Response in Cisco Modeling Labs

Final Capstone Project Report

Course	CN8001 Capstone Project
Platform	Cisco Modeling Labs (CML) hosted on Proxmox
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Submission date	31st July, 2026

Project status: implemented, tested and evidence-backed

Executive Summary

This project implemented and evaluated a lab-scale Zero Trust Network Architecture (ZTNA) in Cisco Modeling Labs. The design removes implicit trust based on network location and requires every cross-segment flow to pass through a policy enforcement point. FW-CORE provides internal segmentation and least-privilege policy enforcement; FW-EDGE protects the perimeter and a screened DMZ; a separate 802.1X and FreeRADIUS proof-of-concept demonstrates certificate-based device admission; a behavior-based IPS automatically quarantines hosts that repeatedly probe restricted zones; and Splunk centralizes application, authentication and system telemetry from the DMZ web server.

Testing confirmed the intended contrast between permitted business traffic and denied lateral movement. Users could access approved web services but could not directly reach the database, storage or administrative tiers. The DMZ server could use tightly limited outbound services but could not initiate connections toward protected internal networks. Repeated denied probes triggered dynamic quarantine, while repeated requests for nonexistent DMZ paths were detected by Splunk. The resulting system demonstrates prevention, detection and response as complementary controls rather than relying on a single perimeter.

Scope boundary: Splunk currently receives Ubuntu-5 Nginx, authentication and system logs. FW-CORE sends ASA deny syslog to the IPS node for automated response. Direct forwarding of FW-CORE/FW-EDGE logs into Splunk was designed as future work and is not presented as completed.

Acknowledgements

The project team would like to thank **Waqas Siddique** for providing guidance, technical feedback and support throughout the development of this project. We also appreciate Toronto Metropolitan University for providing access to Cisco Modelling Labs and the virtual infrastructure required to design, configure and test the proposed network.

We would also like to acknowledge the contributions of all group members. Each member supported the project through network design, firewall configuration, Network Access Control testing, intrusion-prevention development, Splunk integration, attack simulation, troubleshooting and documentation.

Certification of Authorship

We hereby declare that we are the authors of this case study report and that the work presented is original. All external information, documentation and technical material used in this report have been appropriately acknowledged and cited.

We confirm that this report accurately represents the network architecture, configurations, testing procedures and results completed by the project team. No test results or evidence have been intentionally fabricated or misrepresented.

We authorize Toronto Metropolitan University to retain and reproduce this report for academic and scholarly purposes.

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Date : 31st July, 2026

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Evidence convention: Each blue evidence box marks the exact location where a screenshot belongs. Replace the box with the captured image and retain the suggested caption. Evidence S20-S22 belongs to the separate NAC proof-of-concept; S24-S25 use process/listener or saved console evidence if the IPS was not installed as a systemd service.

Introduction

Traditional networks often trust devices once they are inside the perimeter. Zero Trust rejects that assumption. NIST SP 800-207 describes an approach in which no subject or asset receives implicit trust solely because of its physical or network location [1]. In this project, that principle was translated into enforceable network rules, device-verification controls, centralized monitoring and automated containment.

The work was completed in Cisco Modeling Labs hosted on Proxmox. Virtual Cisco ASAv firewalls and IOSvL2 switches provide routing, VLAN segmentation and access control. Ubuntu systems provide application, RADIUS, IPS and logging functions. Splunk Enterprise runs locally on a laptop inside WSL and receives data from a Universal Forwarder on the DMZ web server.

Project Objectives

- Force all inter-zone traffic through a firewall policy enforcement point.
- Apply default-deny controls with narrow, service-specific exceptions.
- Isolate public-facing services in a DMZ behind a separate edge firewall.
- Demonstrate certificate-based device admission and quarantine in a separate NAC lab.
- Detect suspicious network behavior and automatically quarantine the source port.
- Centralize DMZ application and host telemetry in Splunk for detection and reporting.
- Produce repeatable evidence through ACL counters, packet simulation, service logs and dashboard results.

Scope and Assumptions

The project is an educational proof-of-concept rather than a production deployment. The virtual appliances are single instances, management automation uses lab credentials, the IPS uses a tunable behavior threshold instead of deep-packet signatures, and certificate-based NAC was validated in an isolated companion lab. These boundaries do not reduce the value of the experiment; they establish what was actually demonstrated and what remains future work.

1.Background

Traditional network-security models mainly protect the boundary between an organization and the Internet. Once a device enters the internal network, it may receive more access than it requires. This creates a security risk because a compromised internal device can attempt lateral movement toward applications, databases, storage systems and administrative resources.

Zero Trust removes the assumption that internal network location automatically makes a user or device trustworthy. NIST SP 800-207 states that access should be evaluated according to the identity of the subject, the requested resource, device condition and applicable security policy rather than physical or network location alone [1]. Access should therefore be limited to the specific resource and service required for a legitimate task.

Network segmentation supports Zero Trust by separating users, administrators, applications, databases, storage systems and public services into distinct security zones. However, VLAN separation alone does not enforce security. Traffic between those VLANs must pass through a policy enforcement point capable of applying explicit access-control rules. In this project, FW-CORE performs this function for internal traffic, while FW-EDGE protects the Internet boundary and DMZ. Cisco ASA access-control lists provide explicit service-based permits followed by logged default-deny rules [2].

Zero Trust also requires visibility and continuous evaluation. Certificate-based 802.1X verifies a device before granting network access, the behavioural IPS monitors denied network activity and Splunk collects host and application logs for centralized investigation. These controls combine prevention, detection and response rather than relying on a single firewall.

2. Architecture and Addressing

The topology separates Internet access, the DMZ, internal user and server tiers, management services, monitoring and attack simulation. FW-CORE is the internal policy enforcement point. FW-EDGE is the perimeter firewall. VLAN trunks carry tagged segments to the firewalls, while endpoints use access ports in their assigned VLANs.

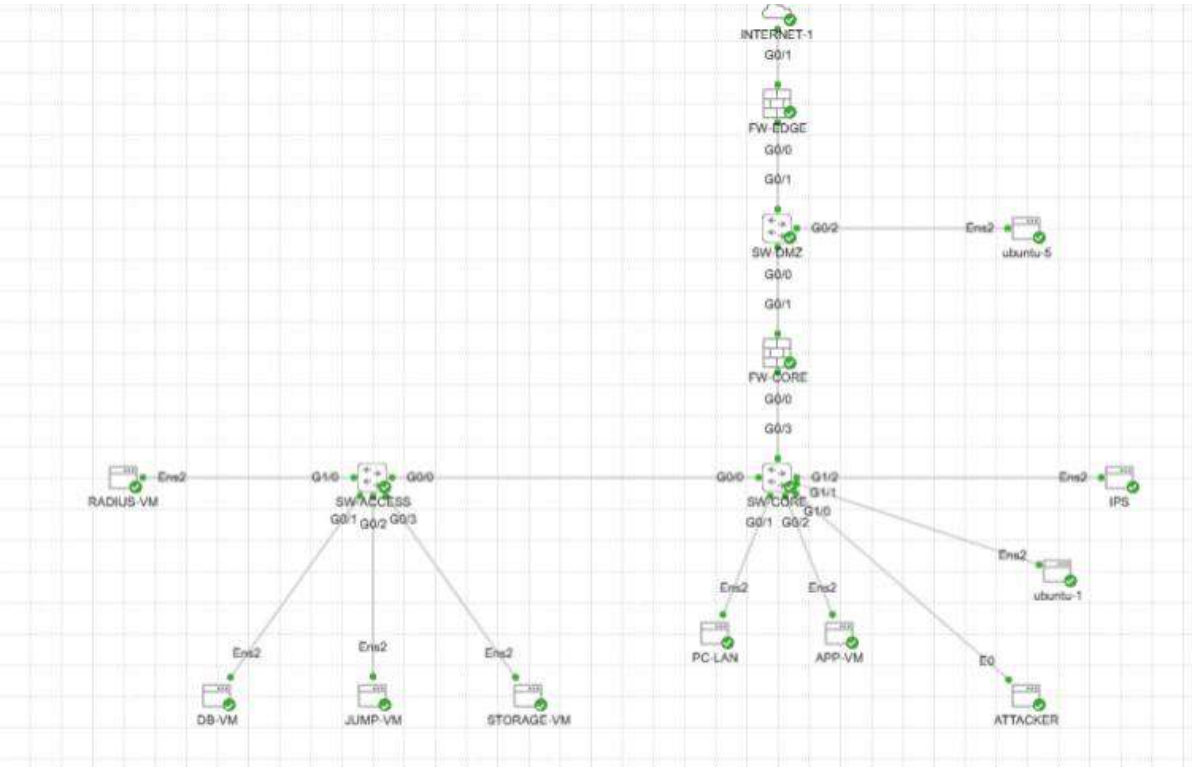


Figure 1. Final Cisco CML Zero Trust topology.

2.1 VLAN and IP Plan

VLAN	Zone / Purpose	Subnet	Gateway	Representative host
20	USER	192.168.20.0/24	192.168.20.1	PC-LAN / Attacker
30	ADMIN	192.168.30.0/24	192.168.30.1	JUMP-VM .50 / IPS .60
40	DC-APP	192.168.40.0/24	192.168.40.1	APP-VM .10
41	DC-DB	192.168.41.0/24	192.168.41.1	DB-VM .20
42	DC-STORAGE	192.168.42.0/24	192.168.42.1	STORAGE-VM .30
50	DMZ	172.16.50.0/24	172.16.50.1	Ubuntu-5 .10
99	QUARANTINE	192.168.99.0/24	192.168.99.1	Isolated hosts
200	EDGE-TRANSIT	172.16.99.0/30	N/A	FW-EDGE .1 / FW-CORE .2

2.2 Switching Design

SW-CORE carries the USER, ADMIN, application, database, storage and quarantine VLANs. SW-ACCESS extends the ADMIN, database and storage VLANs to the corresponding servers. SW-DMZ transports VLAN 50 and the edge-transit VLAN to FW-EDGE while keeping the DMZ server on a dedicated access port.

>_ UBUNTU-5 >_ ATTACKER >_ SW-CORE >_ UBUNTU-1

```
SW-CORE#
SW-CORE#terminal length 0
SW-CORE#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi2/0, Gi2/1, Gi2/2, Gi2/3
20   USER                             active
30   ADMIN                            active    Gi1/2, Gi1/3
40   DC-APP                           active    Gi0/2
41   DC-DB                            active
42   DC-STORAGE                       active
99   QUARANTINE                       active    Gi0/1, Gi1/0, Gi1/1
999  UNUSED-NATIVE                    active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
SW-CORE#show interfaces status

Port      Name               Status       Vlan       Duplex  Speed Type
Gi0/0                        connected    trunk      a-full   auto RJ45
Gi0/1                        connected    99         a-full   auto RJ45
Gi0/2                        connected    40         a-full   auto RJ45
Gi0/3                        connected    trunk      a-full   auto RJ45
Gi1/0                        connected    99         a-full   auto RJ45
Gi1/1                        connected    99         a-full   auto RJ45
Gi1/2                        connected    30         a-full   auto RJ45
Gi1/3                        connected    30         a-full   auto RJ45
Gi2/0                        notconnect   1          a-full   auto RJ45
Gi2/1                        notconnect   1          a-full   auto RJ45
```

S02: SW-CORE VLAN membership, access ports and 802.1Q trunks.

>_ UBUNTU-5 × >_ ATTACKER >_ SW-CORE >_ UBUNTU-1

```
SW-CORE#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Gi0/0       on               802.1q         trunking      999
Gi0/3       on               802.1q         trunking      999

Port        Vlans allowed on trunk
Gi0/0       30,41-42,99
Gi0/3       20,30,40-42,99

Port        Vlans allowed and active in management domain
Gi0/0       30,41-42,99
Gi0/3       20,30,40-42,99

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/0       30,41-42,99
Gi0/3       20,30,40-42,99
SW-CORE#show running-config interface GigabitEthernet0/0
Building configuration...

Current configuration : 216 bytes
!
interface GigabitEthernet0/0
 switchport trunk allowed vlan 30,41,42,99
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
 switchport nonegotiate
 negotiation auto
end
```

S03: SW-ACCESS server VLAN assignments and trunk to SW-CORE.

```
SW-CORE#show running-config interface GigabitEthernet0/3
Building configuration...

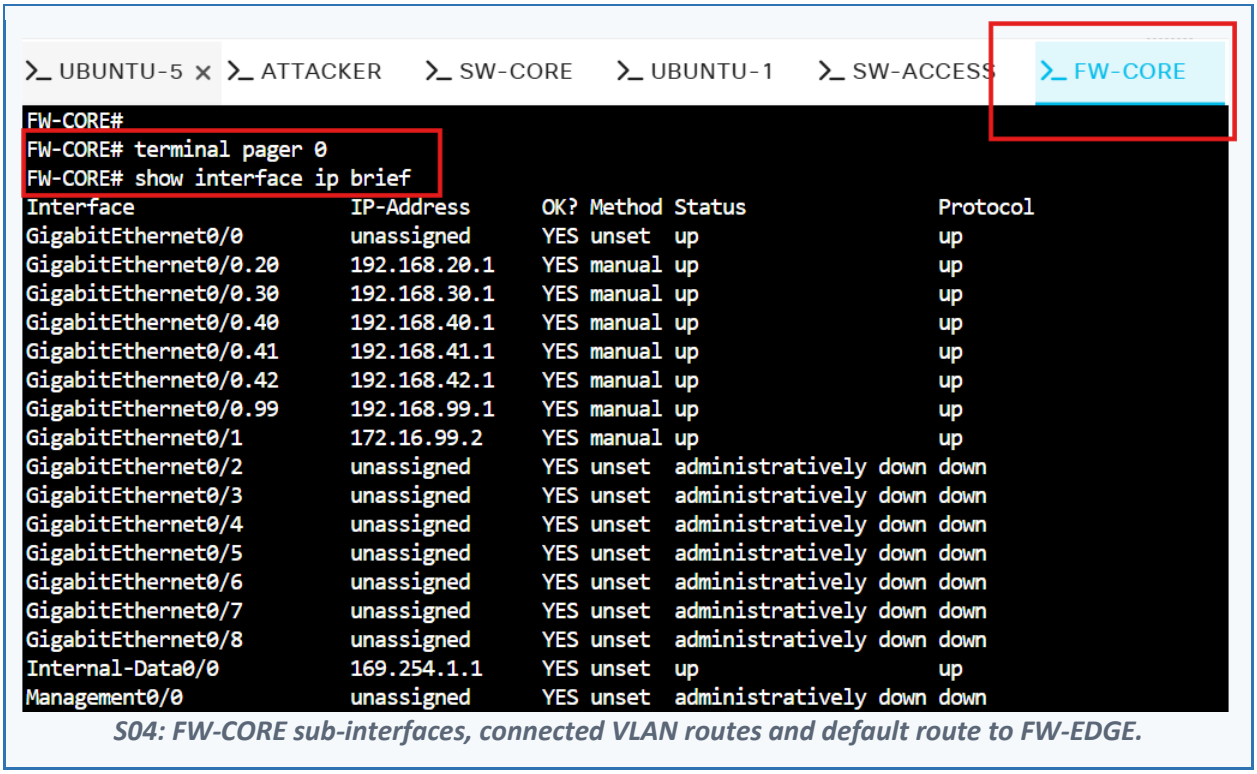
Current configuration : 219 bytes
!
interface GigabitEthernet0/3
 switchport trunk allowed vlan 20,30,40-42,99
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
 switchport nonegotiate
 negotiation auto
end

SW-CORE#
```

S03.1: SW-ACCESS server VLAN assignments and trunk to SW-CORE.

3. Core Zero Trust Enforcement

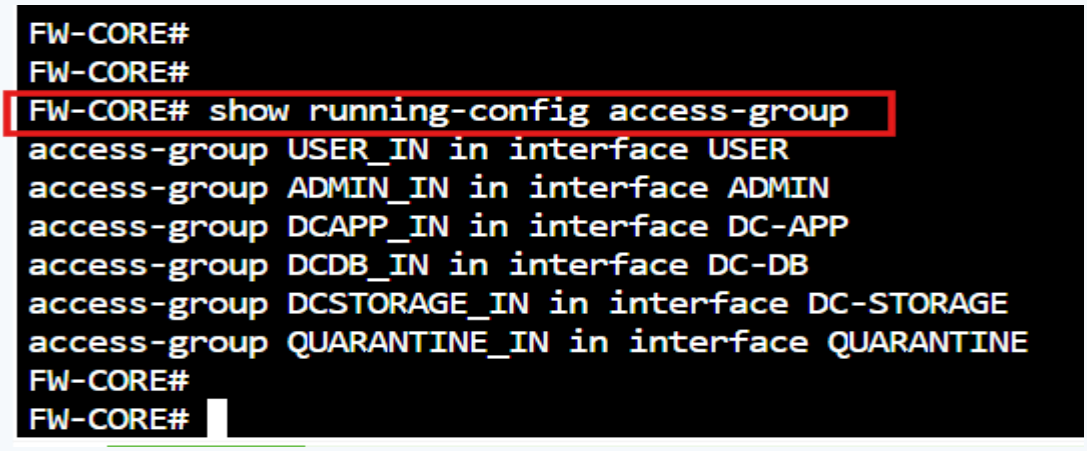
FW-CORE acts as the primary policy enforcement point. Its 802.1Q subinterfaces provide the default gateway for each internal VLAN. Because routing between the VLANs occurs on the firewall, every cross-segment flow is evaluated against an interface-specific ACL. The policy pattern is explicit permit followed by a logged final deny.



S04: FW-CORE sub-interfaces, connected VLAN routes and default route to FW-EDGE.

3.1 Least-Privilege Policy

Source zone	Permitted destination / service	Denied behavior
USER	APP 80/443 and approved ICMP; DMZ 80/443	Direct DB, storage, admin and other traffic
ADMIN	Jump host SSH to APP and DB	Other unsanctioned administrative access
DC-APP	DB 3306; storage 2049/3260 and test ICMP	All other east-west initiation
DC-DB	Approved storage services only	User/admin initiation and unrelated services
DC-STORAGE	Only required return/approved flows	Independent lateral movement



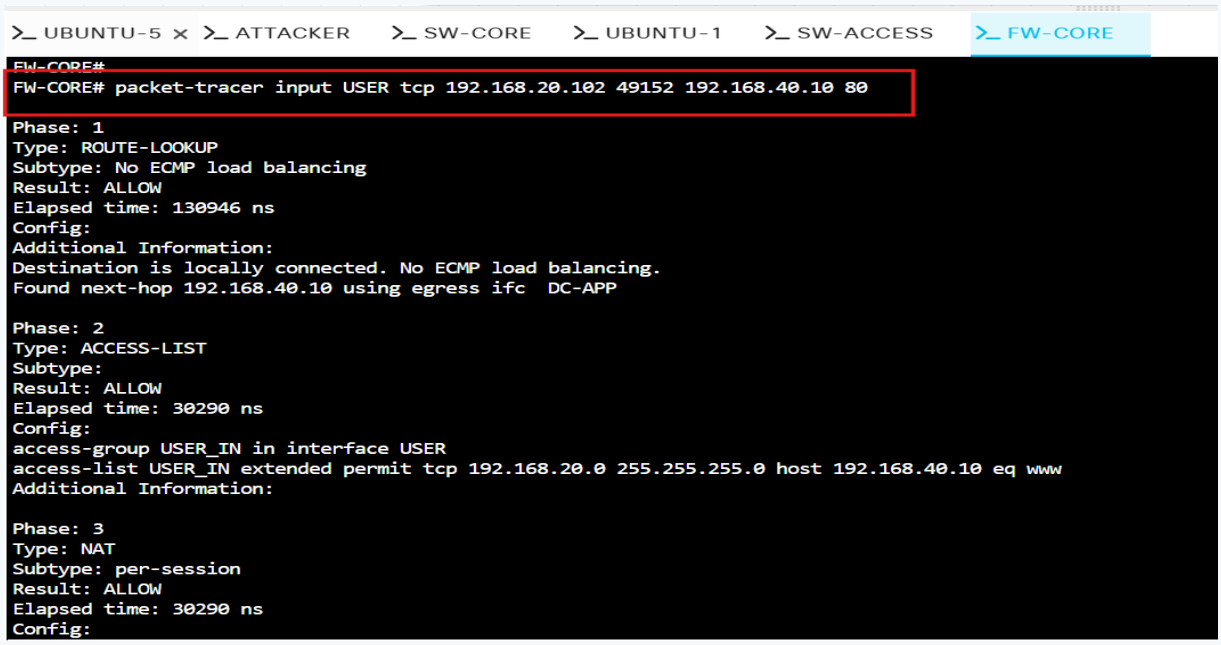
S05: Inbound ACL bindings and USER_IN permits placed above the logged final deny.



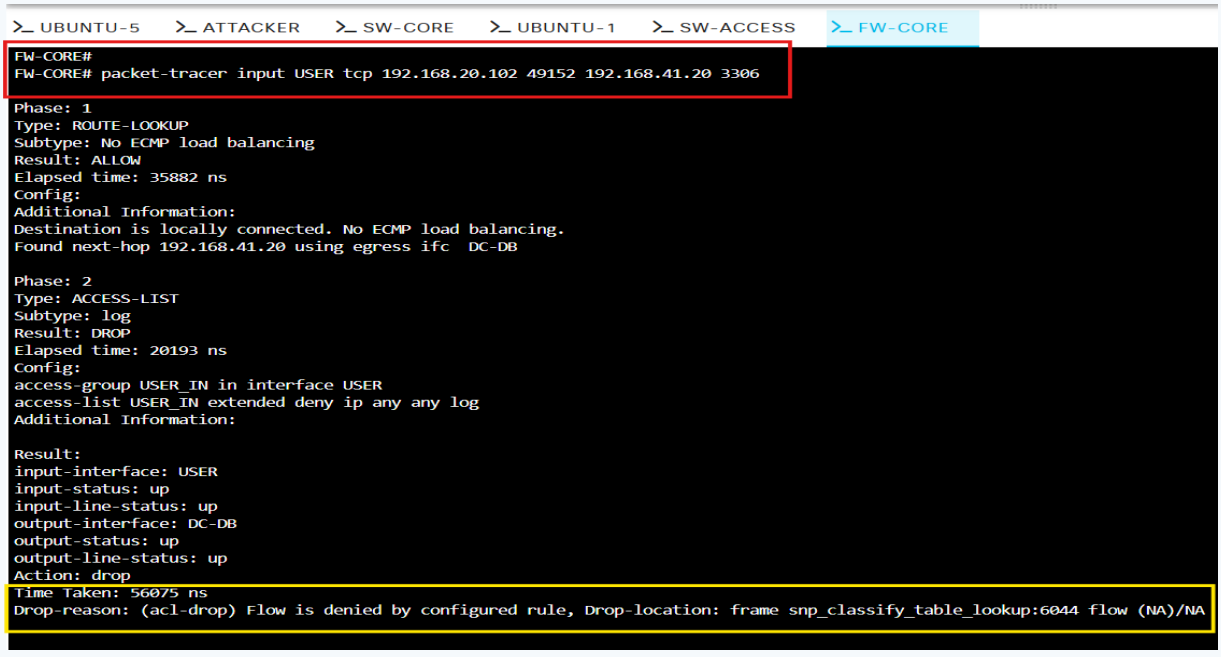
S06: Least-privilege administrative and server-tier ACLs.

3.2 Policy Evaluation Example

A user request to the approved application web service is permitted, while a direct attempt to reach the database is rejected. Cisco ASA packet-tracer validates the complete decision path without modifying live policy.



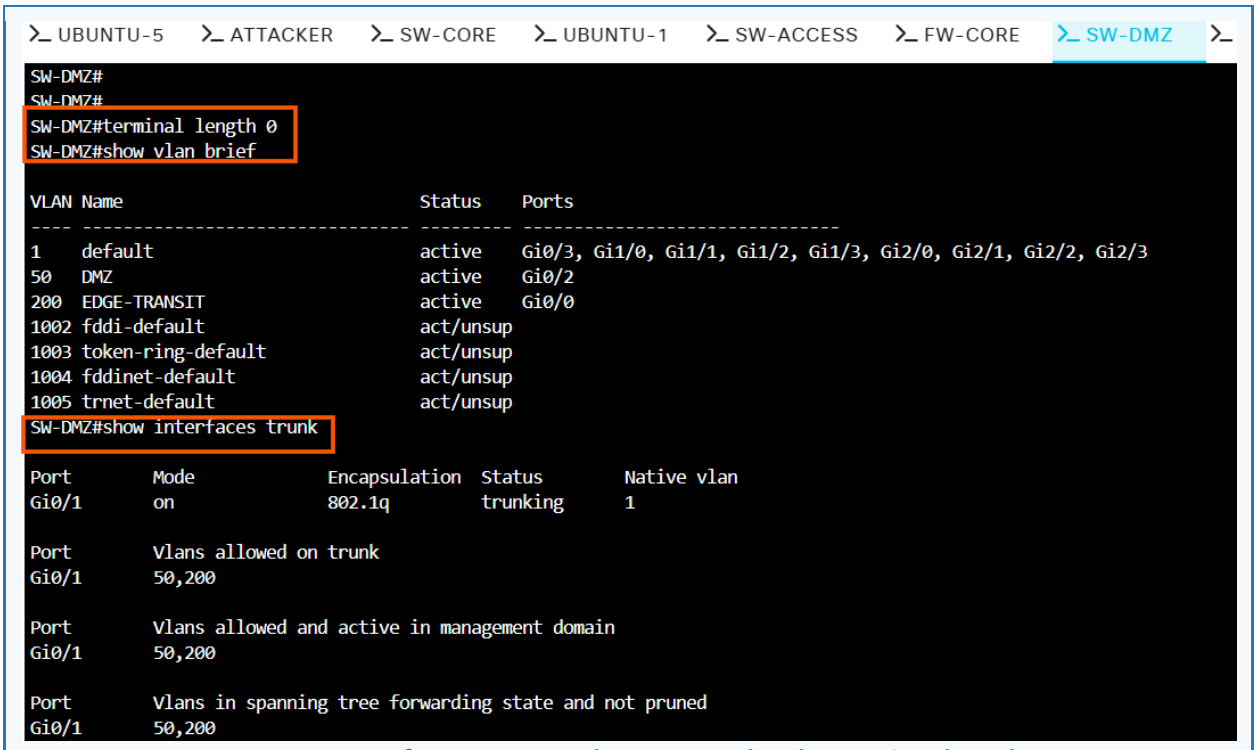
S07: FW-CORE permits the approved APP web path and drops direct USER-to-DB access.



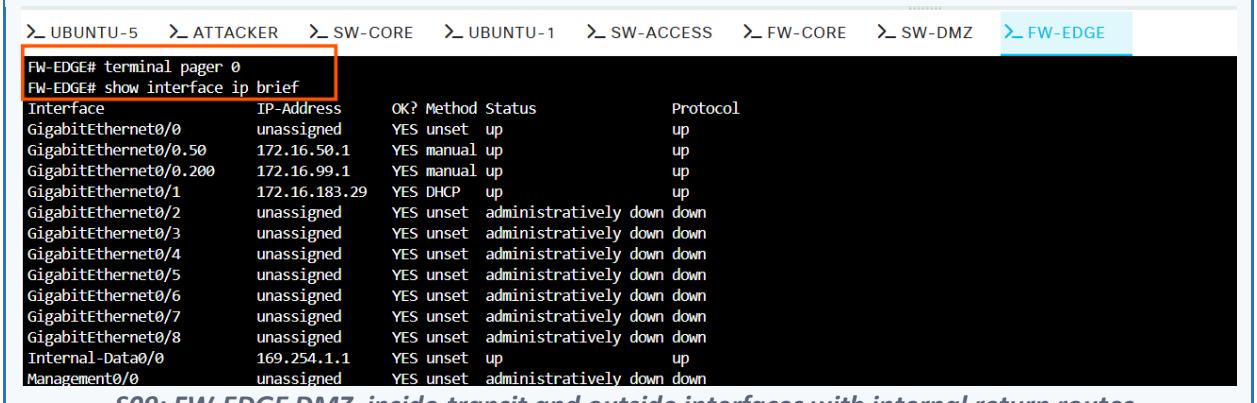
Interpretation: The same source identity receives different outcomes based on the requested resource and service. This is the practical Zero Trust behavior: access is evaluated per transaction, not granted broadly because the source is inside the network.

4. Perimeter Firewall and DMZ

FW-EDGE is inline between the external connector and the internal environment. It terminates the DMZ gateway, the point-to-point transit toward FW-CORE and the DHCP-configured outside interface. Static routes return the internal VLANs through FW-CORE. This screened-subnet design prevents a compromised public web server from directly bypassing firewall inspection.



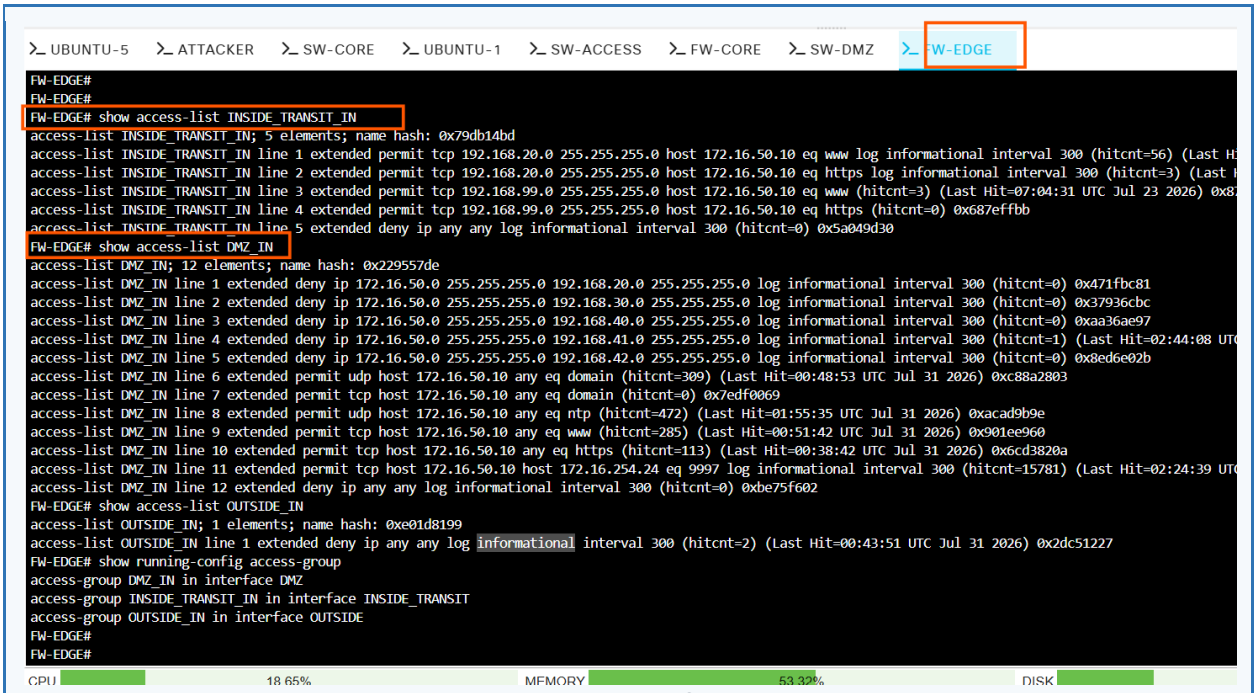
S08: SW-DMZ access ports for VLAN 50 and VLAN 200 plus the restricted trunk to FW-EDGE.



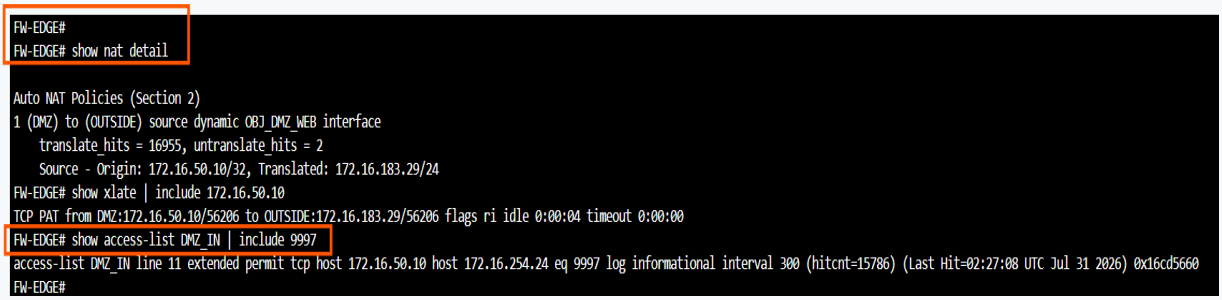
S09: FW-EDGE DMZ, inside-transit and outside interfaces with internal return routes.

4.1 DMZ Policy and NAT

Users are allowed to access the DMZ web server only on HTTP and HTTPS. DMZ-originated attempts toward each internal subnet are explicitly denied and logged. Ubuntu-5 receives only the minimum outbound services needed for DNS, NTP, web updates and Splunk forwarding. Dynamic PAT translates the private DMZ address when traffic exits the outside interface.



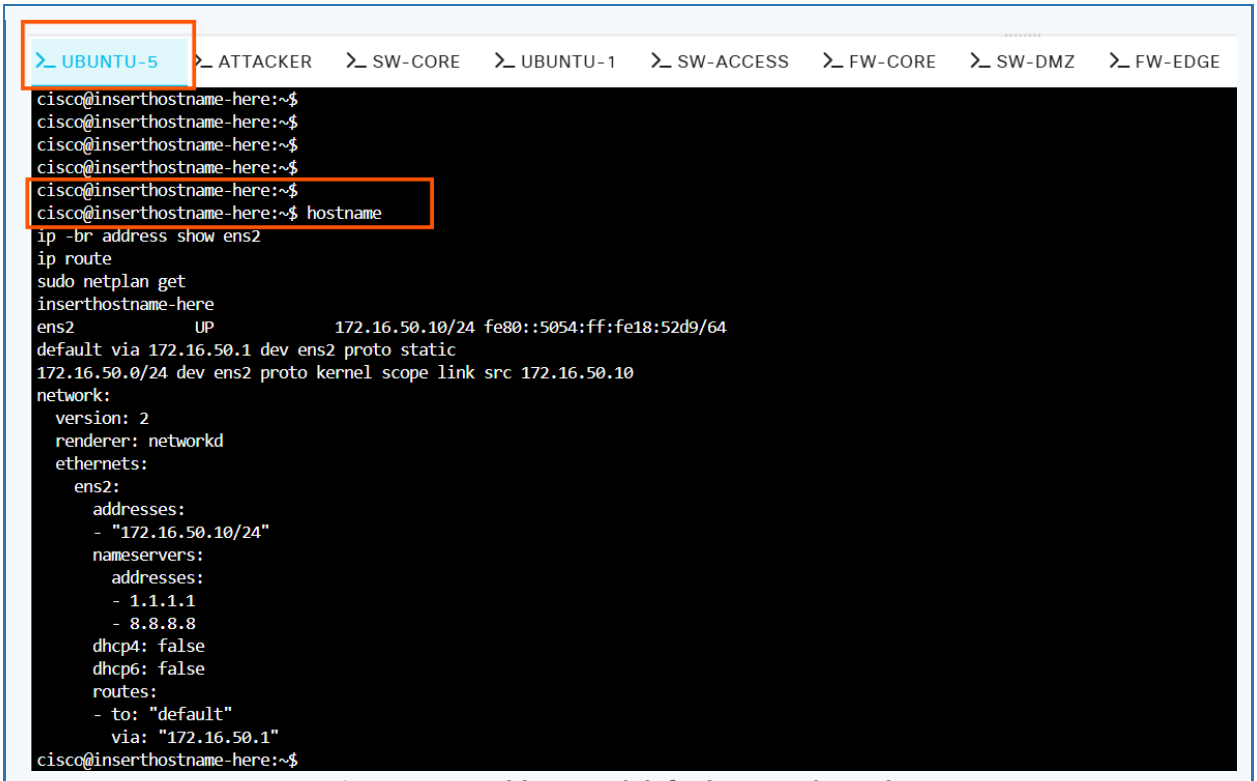
S10: FW-EDGE least-privilege DMZ policy, final denies and PAT translations.



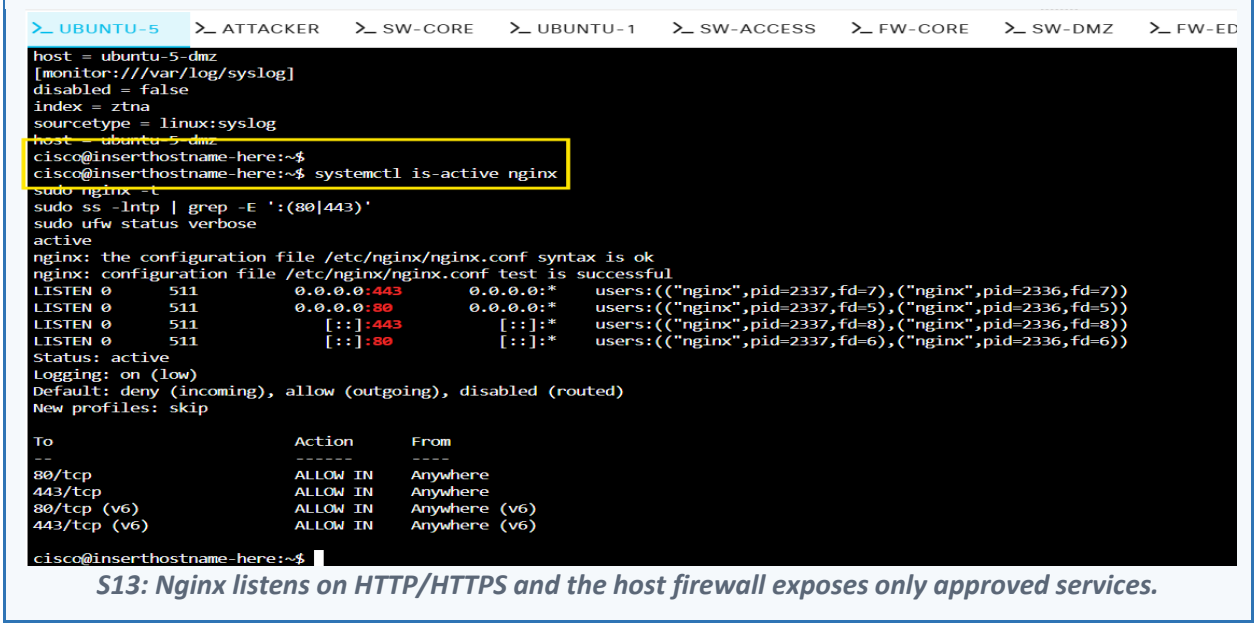
S11: Only Ubuntu-5 is permitted to open TCP 9997 toward the Splunk receiver.

4.2 Ubuntu-5 DMZ Web Server

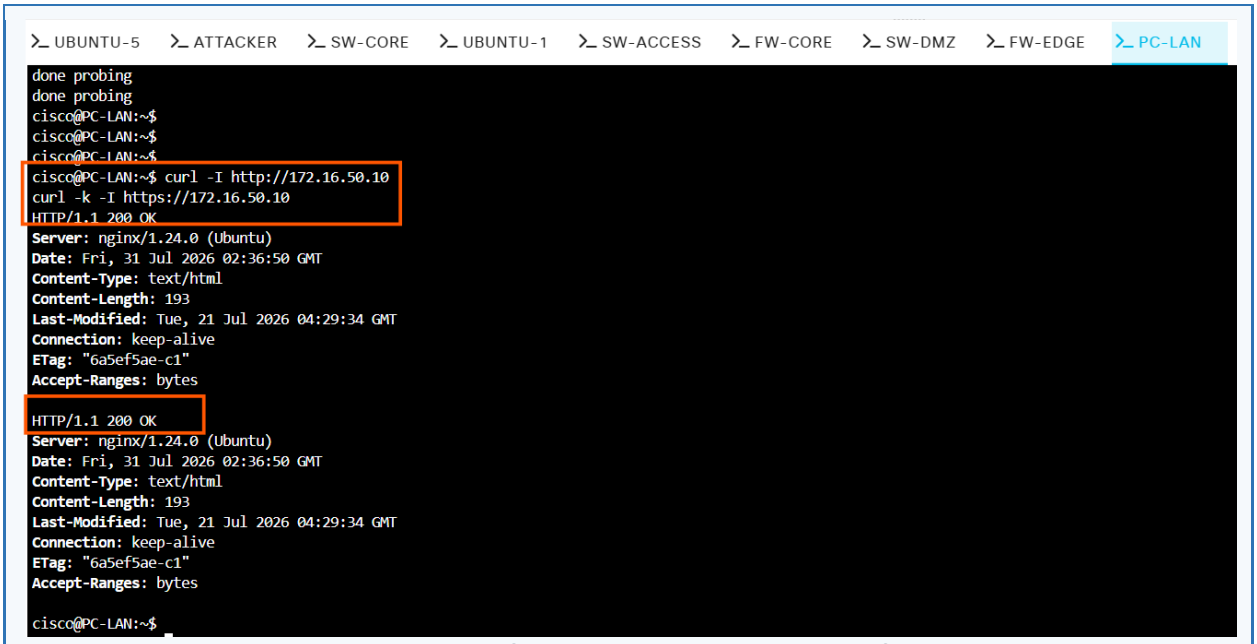
Ubuntu-5 uses the persistent address 172.16.50.10/24 with FW-EDGE 172.16.50.1 as its default gateway. Nginx serves HTTP and HTTPS. UFW denies unsolicited inbound traffic by default and exposes only the intended web ports; administrative access is limited to the approved management path.



S12: Persistent DMZ address and default route through FW-EDGE.



S13: Nginx listens on HTTP/HTTPS and the host firewall exposes only approved services.



S14: The DMZ server returns successful HTTP and HTTPS responses from an authorized client.

5. Network Access Control Proof-of-Concept

Important scope note: The 802.1X/FreeRADIUS experiment was completed and verified in a separate isolated NAC lab. It demonstrates the device-verification pillar but is not shown as an inline component of the final main topology.

SW-NAC acts as the 802.1X authenticator and FreeRADIUS as the authentication server and certificate authority. A client presenting a valid EAP-TLS certificate receives VLAN 10 (CORP). A device without a valid certificate is rejected or placed into VLAN 99 (QUARANTINE) using the IOSvL2 legacy guest/auth-fail VLAN behavior.

1. The endpoint begins 802.1X authentication before normal Layer 3 access.
2. FreeRADIUS validates the client certificate against the lab certificate authority.
3. An Access-Accept includes the tunnel attributes that assign the corporate VLAN.
4. Missing or invalid certificate material produces EAP failure and quarantine.

6. Behavioral IPS and Automated Quarantine

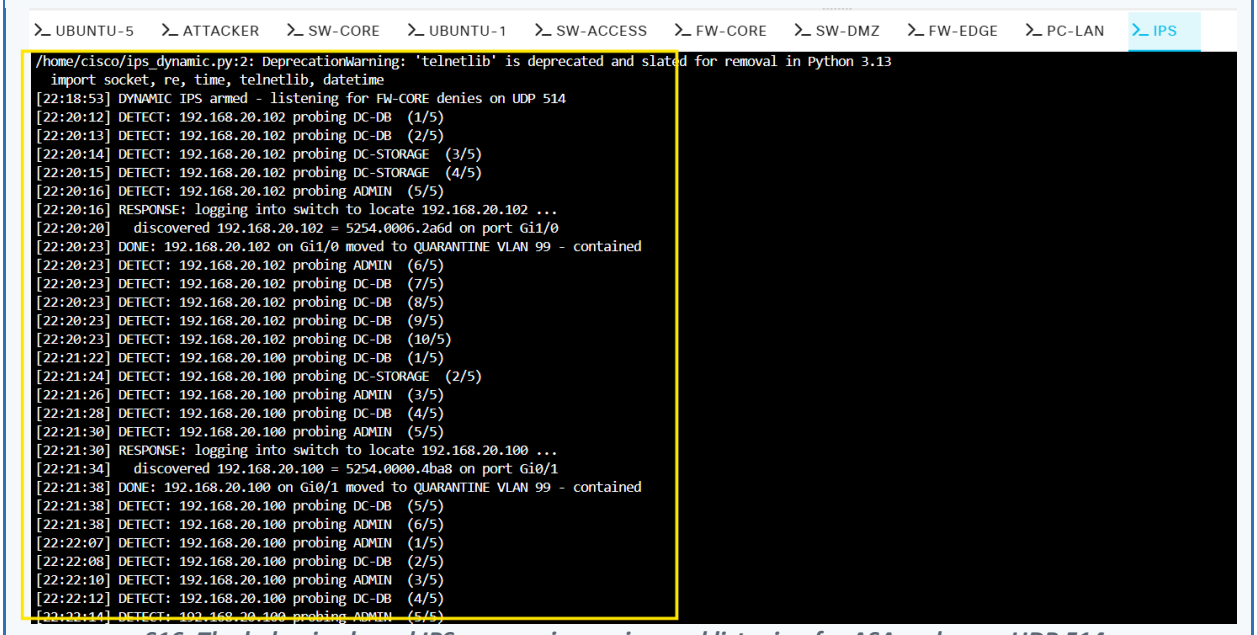
The IPS node resides at 192.168.30.60 in the protected ADMIN VLAN. FW-CORE sends informational syslog messages to the node over UDP 514. The key event is Cisco ASA message 106100, generated when a logged ACL entry denies a flow [2][3]. The Python service correlates behavior rather than inspecting packet payloads.

If one source creates five or more denied attempts toward restricted zones within fifteen seconds, the IPS identifies the source switch port dynamically. It resolves IP to MAC, MAC to switch port and then

moves that access port into VLAN 99. The response was proven against both a dedicated attacker and a normal user host, demonstrating that behavior—not a trusted label—drives containment.



S15: FW-CORE logging is enabled and forwards informational events to IPS 192.168.30.60.



S16: The behavior-based IPS process is running and listening for ASA syslog on UDP 514.

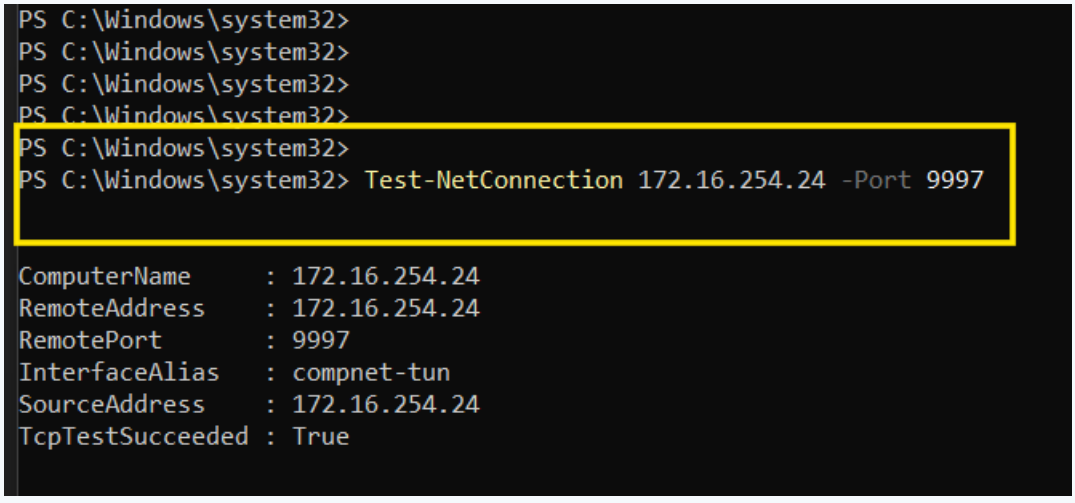
6.1 IPS Capabilities and Limits

Demonstrated capability	Known limitation
Detects rapid port scans and cross-segment probing that produces firewall denies.	Does not inspect payloads or replace a signature-based IDS/IPS.
Responds automatically by moving the source port to VLAN 99.	A very slow scan below the threshold may evade the detector.
Dynamically discovers the switch port; no attacker-port hardcoding.	Requires reliable ARP/MAC/port mapping and protected switch-management access.
Applies the same response to trusted and untrusted hosts.	Attacks over an already permitted service require application/SIEM detection.

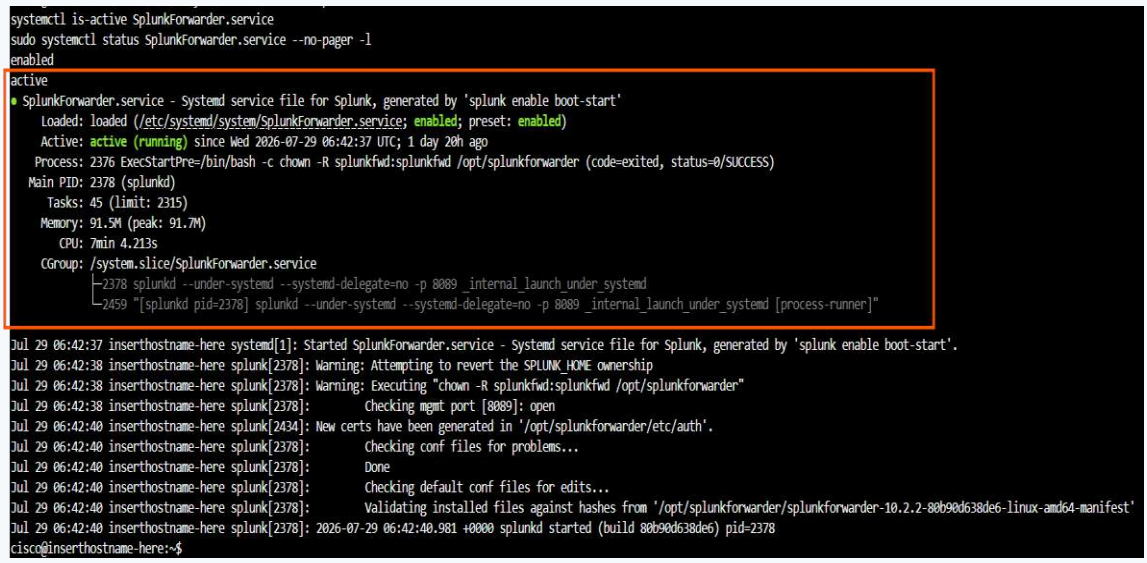
7. Splunk SIEM Integration

Splunk Enterprise runs in Ubuntu WSL on the project laptop. Because the CML lab reaches the laptop through a university VPN adapter, Windows port proxy publishes 172.16.254.24:9997 and forwards the connection to the local Splunk receiver. The Universal Forwarder on Ubuntu-5 sends events through FW-EDGE using the single least-privilege TCP 9997 exception.

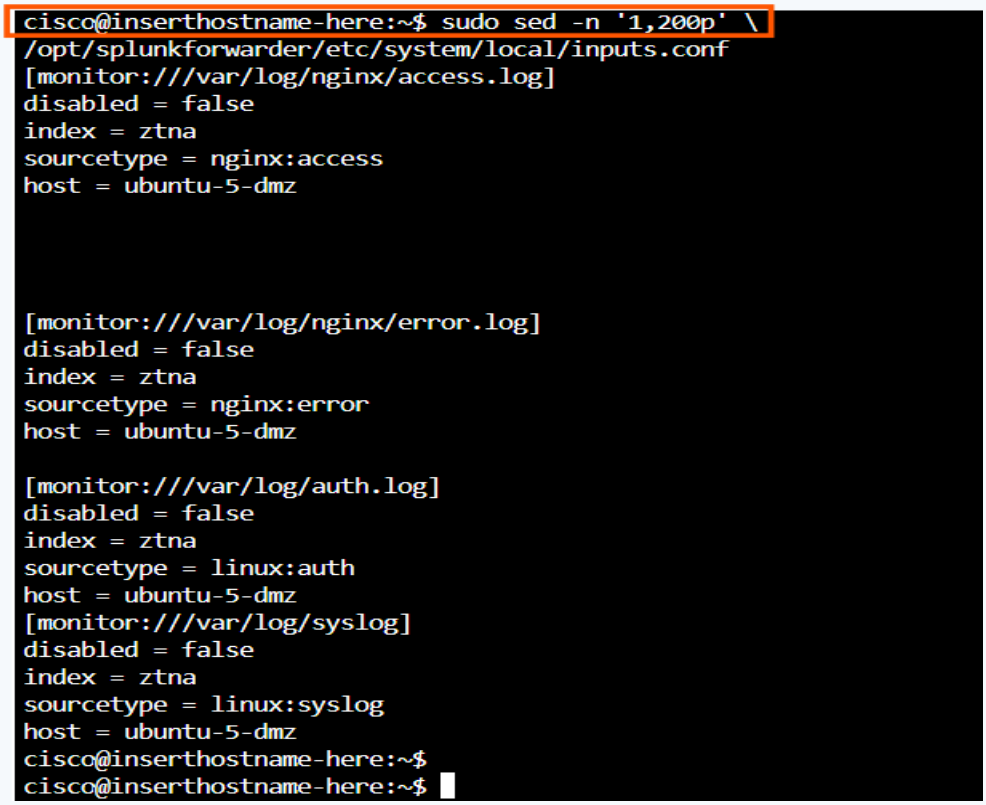
The Universal Forwarder monitors files with inputs.conf and sends them to the configured receiver, consistent with Splunk's documented file-monitor and forwarding model [4][5]. Events are stored in the dedicated ztna index with host=ubuntu-5-dmz.



S17: Windows port proxy and the Splunk TCP 9997 receiver are reachable.



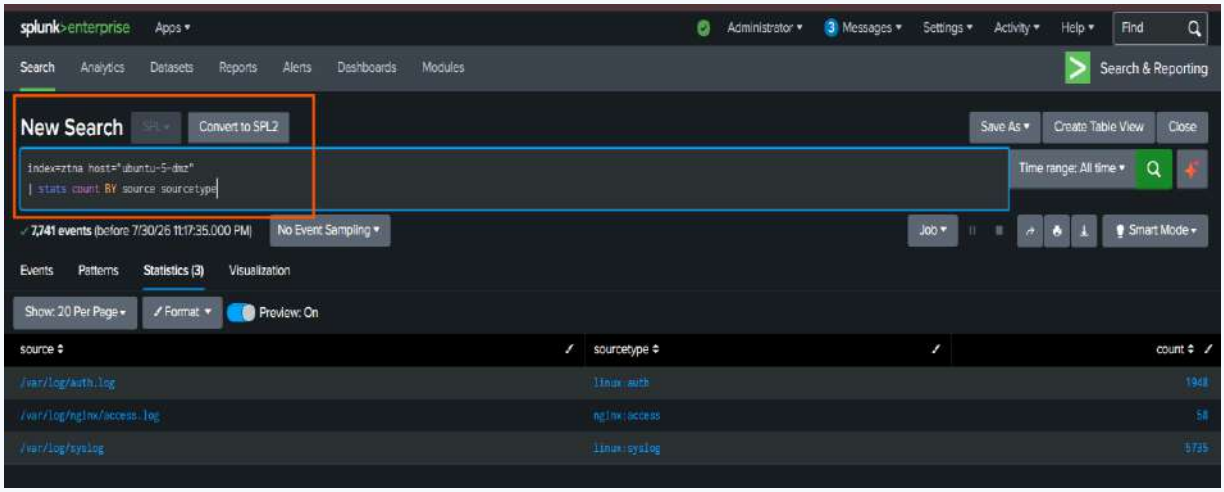
S18: The forwarder is enabled, active and connected to 172.16.254.24:9997.



S19: Ubuntu-5 forwards Nginx access/error, authentication and system logs into index ztna.

7.1 Collected Telemetry

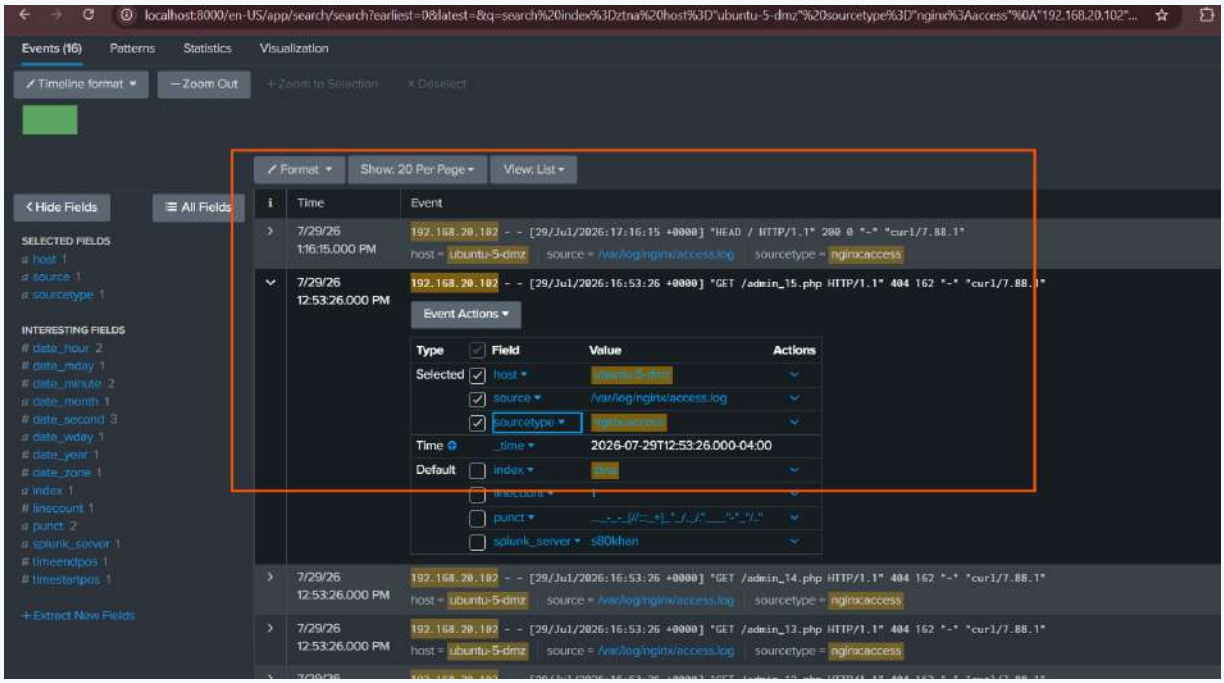
Log source	Sourcetype	Security value
/var/log/nginx/access.log	nginx:access	Client requests, paths, status codes and user agents
/var/log/nginx/error.log	nginx:error	Web-service errors and operational anomalies
/var/log/auth.log	linux:auth	Authentication failures, sudo activity and session events
/var/log/syslog	linux:syslog	System, service and reboot/persistence events



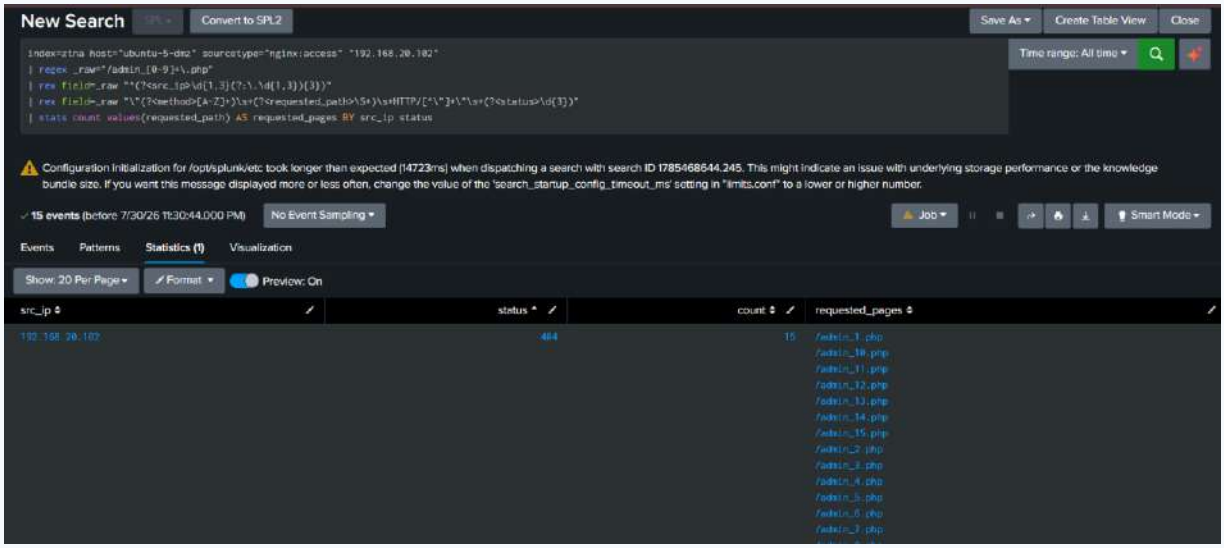
S20: Splunk receives events from all four configured Ubuntu-5 log sources.

7.2 Detection: Repeated DMZ 404 Requests

The attacker at 192.168.20.102 requested fifteen nonexistent paths named /admin_1.php through /admin_15.php. The requests were permitted because HTTP to the DMZ is an approved path, but Nginx returned 404 responses. Splunk grouped the failures by source and detected the repeated enumeration pattern.



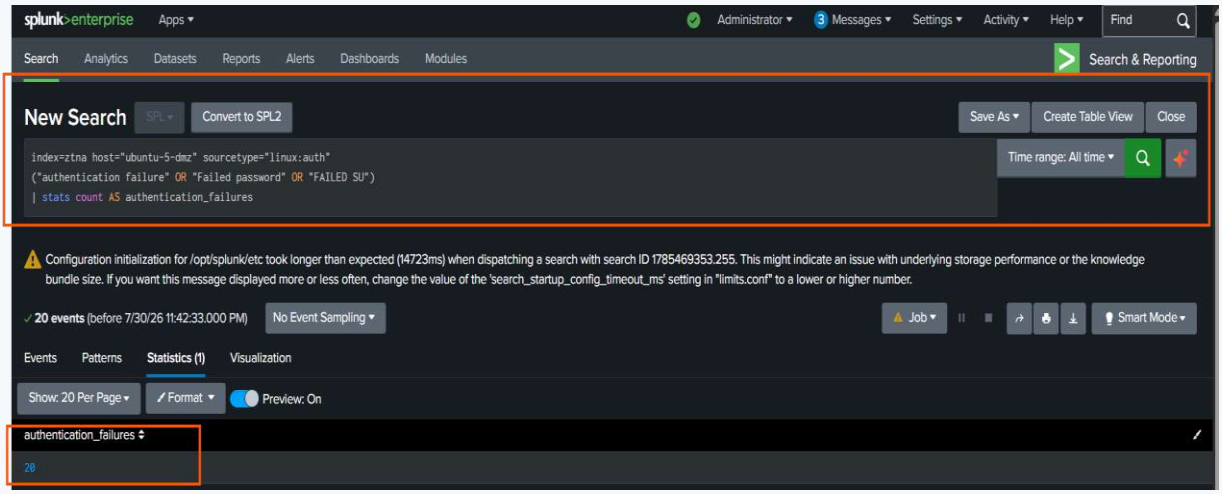
S21: Raw Nginx events show the attacker IP, requested admin paths and HTTP 404 results.



S22: Splunk correlates repeated 404 requests by source IP and lists the probed paths.

7.3 Detection: Repeated Authentication Failures

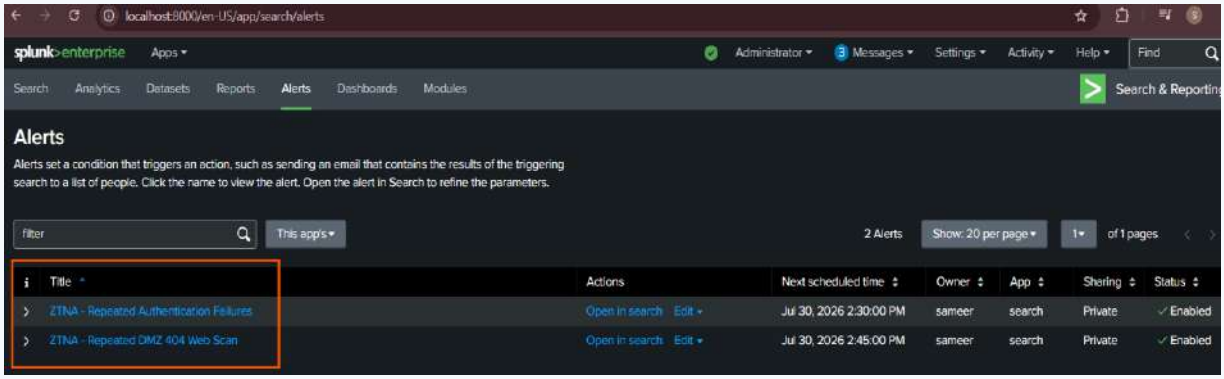
A second scheduled search analyzes /var/log/auth.log for repeated authentication failure patterns. Five or more failures in the configured time window produce a detection result. Splunk provides visibility and alerting; enforcement remains the responsibility of the firewalls, UFW and IPS response logic.



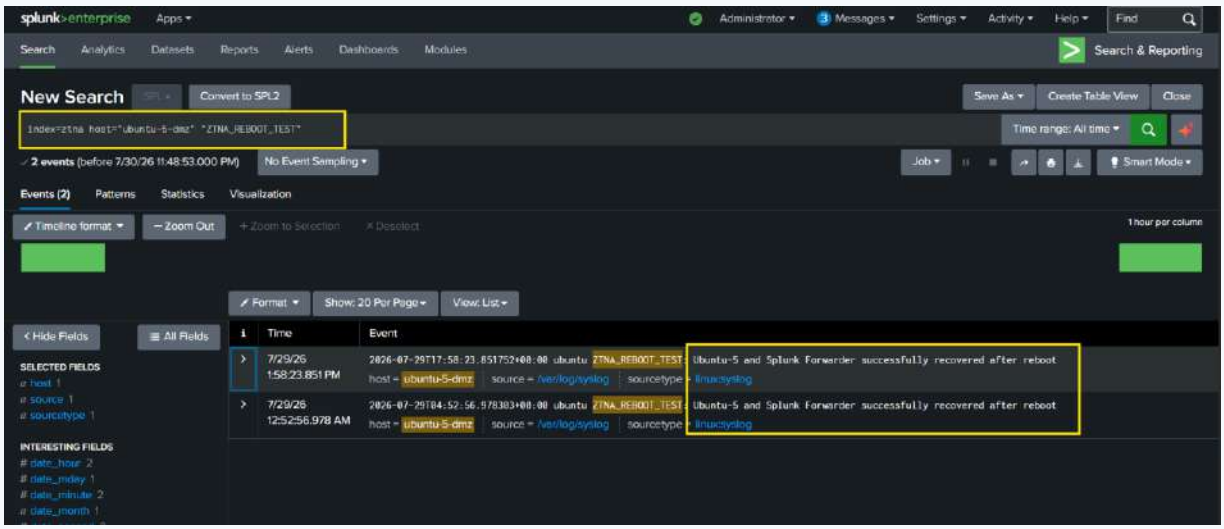
S23: Splunk counts repeated authentication failures on Ubuntu-5.

7.4 Dashboard and Persistence

The ZTNA Security Monitoring Dashboard presents events by log type, suspicious DMZ 404 activity and authentication failures. The forwarder runs under the least-privileged splunkfwd account as an enabled systemd service. A reboot test produced a ZTNA_REBOOT_TEST syslog event in Splunk, proving that networking, Nginx, the forwarder and the end-to-end connection recovered.



S24: Central dashboard for DMZ event volume, suspicious web enumeration and authentication failures.



S25: The post-reboot test event confirms automatic recovery and persistent log forwarding.

8.Alternative Designs

Several alternative designs were considered before the final architecture was selected.

Perimeter-Only Firewall

The simplest alternative was to use one perimeter firewall while allowing unrestricted communication between internal systems. This design would require fewer configurations and network devices, but it would not prevent lateral movement after an internal endpoint became compromised. It was rejected because it did not meet the project's default-deny and least-privilege objectives.

Single Firewall for Internal Segmentation and DMZ

Another option was to connect all internal VLANs and the DMZ to a single firewall. This would reduce the number of ASAv nodes and simplify routing. However, internal segmentation and Internet-facing services would depend on the same firewall instance. A configuration error or appliance failure could affect both security layers.

The selected design uses FW-CORE for internal policy enforcement and FW-EDGE for perimeter and DMZ protection. Although this introduces additional routing and configuration requirements, it provides clearer separation of responsibility and stronger defence in depth.

Signature-Based IDS/IPS

Snort or Suricata could have been deployed to inspect packet payloads and identify known attack signatures. This would provide better visibility into attacks travelling over permitted HTTP or HTTPS paths. However, it would require additional interface placement, traffic mirroring, signature management and performance tuning.

The implemented IPS instead uses FW-CORE deny logs to detect rapid lateral-movement behaviour. It is simpler to demonstrate in CML and can automatically quarantine the source port. Its limitation is that it does not inspect payloads or detect every attack travelling through an already permitted service.

Integrated and Separate NAC Designs

Certificate-based NAC could have been inserted directly into the main topology. This would provide identity-based onboarding for production endpoints, but it could also interrupt the already working VLAN, firewall and IPS configuration.

The project therefore validated EAP-TLS and dynamic VLAN assignment in a separate NAC proof-of-concept. This safely demonstrated device verification without modifying the completed main topology. Full NAC integration remains future work.

9. Measurement Results

Testing used both positive and negative cases. A successful Zero Trust result is not simply that traffic is blocked; required business traffic must still work while unauthorized paths fail and produce evidence.

Test	Expected result	Observed evidence	Outcome
USER -> APP 80/443	Permit	Packet trace / service response	Pass
USER -> DB 3306	Deny and log	USER_IN final deny / packet trace	Pass
JUMP -> DB SSH	Permit	ADMIN_IN hit counter / session	Pass
APP -> DB 3306	Permit	DCAPP_IN permit	Pass
DMZ -> internal VLANs	Deny and log	DMZ_IN explicit denies	Pass
USER -> DMZ HTTP/HTTPS	Permit	Nginx response and access log	Pass
DMZ -> Splunk 9997	Permit only Ubuntu-5	ACL hit, PAT and active forward	Pass
Rapid denied probing	IPS quarantine to VLAN 99	IPS console and switch state	Pass
Repeated DMZ 404s	Splunk detection	Raw events and correlated result	Pass
Ubuntu-5 reboot	Forwarding recovers	ZTNA_REBOOT_TEST in Splunk	Pass
Valid EAP-TLS cert (separate NAC lab)	VLAN 10	RADIUS accept / SW-NAC VLAN	Pass
Unknown NAC device (separate lab)	VLAN 99	RADIUS reject / SW-NAC VLAN	Pass

9.1 End-to-End Attack Narrative

1. A host in VLAN 20 begins as an ordinary internal endpoint.
2. It attempts lateral movement toward the database, storage or administrative zones.
3. FW-CORE blocks the requests because USER_IN contains no matching permit.
4. The logged ASA denies are sent to the protected IPS node.
5. The IPS correlates the burst, discovers the source switch port and moves it to VLAN 99.
6. If the host instead enumerates the approved DMZ web service, Nginx records the requests and Splunk detects the repeated 404 pattern.
7. The combined design therefore prevents unauthorized network access, detects suspicious behavior on denied and allowed paths, and contains the offending endpoint.

10. Mapping to NIST SP 800-207

Zero Trust principle	Project implementation
All resources are treated as resources.	Users, applications, databases, storage, the DMZ server and management services are placed in explicit zones.
Communication is secured regardless of location.	Cross-segment traffic is inspected by FW-CORE or FW-EDGE; local location does not bypass policy.
Access is granted per session and with least privilege.	ACLs allow only required host, destination and service combinations; final denies block the remainder.
Access is determined by dynamic policy.	The IPS changes switch-port authorization based on observed behavior; NAC assigns VLAN based on certificate result.
The enterprise monitors asset behavior.	Firewall counters, IPS correlation and Splunk dashboards provide continuous evidence.
Authentication and authorization occur before access.	The separate NAC lab validates EAP-TLS certificate identity before corporate VLAN assignment.
Telemetry improves policy and response.	Nginx, Linux and firewall events support detection, investigation and automated quarantine.

11. Post Analysis

11.1 Strengths

- Defense in depth: internal segmentation, a separate edge firewall, host firewalling, behavioral response and centralized monitoring.
- Least privilege: narrow service-based permits instead of broad subnet trust.
- Reduced blast radius: database and storage tiers cannot be reached directly from the user zone.
- Consistent enforcement: trusted hosts can still be quarantined when behavior violates policy.
- Evidence-driven validation: permit/deny outcomes are supported by counters, packet traces, service logs and Splunk events.
- Operational persistence: key firewall, switch, Nginx and Universal Forwarder configurations survive restart when saved/enabled.

11.2 Limitations

- The lab uses single firewall, switch, IPS and Splunk instances; they are potential single points of failure.
- The behavior-based IPS is not a deep-packet inspection engine and does not detect every payload exploit.
- The threshold can miss low-and-slow scans or create false positives if poorly tuned.
- NAC was validated in a separate lab and has not yet been merged into the main topology.

- Splunk currently observes Ubuntu-5 and Nginx/Linux activity, not every ASA event from FW-CORE and FW-EDGE.
- The Windows VPN/port-proxy path introduces a laptop dependency for real-time Splunk ingestion.
- Virtual ASAv license warnings and CML image behavior differ from production appliances.

12. Lessons Learned

1. ACL order is critical. A final deny placed above a permit shadows it; permits must precede the catch-all deny.
2. Management-plane reachability matters. RADIUS, syslog and Splunk forwarding fail if the control-plane path is missing even when data-plane VLANs appear correct.
3. FreeRADIUS certificate paths and file permissions must match the service account.
4. Virtual IOSvL2 images may require legacy 802.1X guest-vlan and auth-fail syntax.
5. Persistent Linux addressing belongs in netplan; temporary ip commands disappear after reboot.
6. CML Stop/Start preserves disk state, while Wipe erases node configuration.
7. Layer 2 authentication occurs before ordinary IP communication, which explains why an unauthorized NAC client cannot use a static address to bypass control.
8. A monitoring system is useful only when time zone, input files, forwarder startup and receiver reachability are verified.
9. Negative testing is as important as positive testing: a valid architecture proves both what works and what is intentionally blocked.

13. Future Work

The next highest-value improvement is centralized ASA logging in Splunk. FW-CORE already generates and forwards message 106100 to the IPS. A dedicated Ubuntu collector on the ADMIN VLAN could receive a second syslog stream, write the events to /var/log/asa.log and forward that file to Splunk with sourcetype cisco:asa. This would preserve the working IPS path while adding SIEM visibility.

- Add a dedicated ADMIN-VLAN syslog collector and Universal Forwarder for FW-CORE/FW-EDGE events.
- Install the Splunk Add-on for Cisco ASA and build source, destination, port, ACL and zone dashboards.
- Use encrypted forwarder-to-indexer communication and replace the temporary laptop port-proxy dependency.
- Integrate the certificate-based NAC flow into the main topology after validating VLAN and routing interactions.
- Add Snort or Suricata for payload-aware signatures on approved paths such as HTTP/HTTPS.
- Harden switch automation with SSH/API credentials, command authorization and audited service accounts.

- Deploy redundant firewalls, switching and monitoring components to remove single points of failure.
- Tune IPS thresholds using baseline traffic and add controlled release/recovery from VLAN 99.

Safe integration principle: Add telemetry destinations and narrow forwarding permits without removing existing ACLs, changing their order, or replacing the working IPS syslog destination. Validate every addition with read-only show commands and a dedicated attacker before saving.

Conclusion

The project achieved its central objective: internal network location no longer grants broad trust. FW-CORE enforces least-privilege segmentation across user, administration, application, database and storage zones. FW-EDGE places the public web server in a screened DMZ and blocks inward pivoting. The companion NAC lab verifies device identity with EAP-TLS, the IPS converts repeated denied behavior into automatic VLAN quarantine, and Splunk detects suspicious activity that occurs over an otherwise permitted web path.

Together, these controls form a coherent prevention-detection-response loop. The firewalls prevent unauthorized flows, the monitoring systems preserve and correlate evidence, and the IPS contains a misbehaving endpoint. The design remains a lab proof-of-concept with clear limitations, but it demonstrates the practical value of Zero Trust principles and provides a strong foundation for future integration, hardening and high availability.

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Appendix. Configuration Summary

Component	Implemented role	Key configuration
FW-CORE	Internal PEP and inter-VLAN gateway	802.1Q subinterfaces; per-zone ACLs; default route to 172.16.99.1; syslog to IPS
FW-EDGE	Perimeter/DMZ firewall	DMZ and transit subinterfaces; static return routes; DMZ/inside/outside ACLs; PAT
SW-CORE/SW-ACCESS	Segmentation and access switching	VLANs 20/30/40/41/42/99; controlled trunks; quarantine port movement
SW-DMZ	DMZ and transit transport	VLAN 50 and 200; restricted trunk; separate access ports
Ubuntu-5	DMZ web and telemetry source	Static DMZ IP; Nginx 80/443; UFW; Universal Forwarder
IPS node	Behavioral correlation and response	UDP 514 syslog; 5 denies/15 seconds; dynamic IP-MAC-port lookup; VLAN 99
Splunk Enterprise	Central SIEM for DMZ host/application	Index ztna; TCP 9997; alerts and dashboard
Separate NAC lab	Certificate-based device verification	FreeRADIUS EAP-TLS; VLAN 10 success; VLAN 99 failure/quarantine